Intro to Exploratory Data Analysis

Data → A piece of information

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Analyse the data ⇝ Data Analysis

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- To explain the data
- To understand the data

Why we do EDA?

Ans → Helps in solving multiple business prob.

Helps in making predictions using ML

Ex- E-Commerce Platform

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The owner wants to grow the business → analyse data

How to do EDA?

- SQL
- Excel
- Tableau / Power BI Tools
- Programming Language ⇝ [Python, R, etc]

Python ⇝ programming language

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Applications :-

1. Game Development
2. Web Development
3. Automation
4. Data Science (AI/ML)

④ Data Science (AI/ML)

How we use python for EDA?

Basic python programming

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If-else, loops, def, class, dict, list, tuples, etc. } ⇝ Web development / Game development

Advanced Python → Python Libraries

↳ Numpy
↳ Pandas ⇝ EDA
↳ Matplotlib
↳ Seaborn

Numpy :- Used for mathematical computation

↳ It provides fastest computation

Pandas :- Most important for EDA

↳ used for data manipulation (modifications)

Matplotlib / Seaborn →

↳ Data Visualization libraries

↳ Provides a variety of plots / graphs

Line Plot

Bar Chart

Histogram

Piechart

Scatterplot, etc.

Types of Data :-

(1) Numeric Data → consists of only numbers

(2) Categorical Data → Consists of nos., characters, etc

Steps for EDA :-

1. Data Cleaning
2. Data Visualization
3. Data Manipulation

What does the data look like?
Are there any missing values?
Are there extreme values (outliers)?
Which columns are categorical?
" " " numerical?